



TRAINING PROGRAMME

Switching, Routing & Wireless Essentials (CCNA 2)

Design and secure a switched campus network with redundancy and wireless access

42 HOURS

FOR Participants who completed CCNA 1 or have equivalent networking experience

PREPARES FOR CCNA (200-301 v1.1), course 2 of 3



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Overview

This programme is the second of three courses that prepare the Cisco CCNA 200-301 exam, aligned with Cisco Networking Academy Switching, Routing and Wireless Essentials v7. Participants segment networks with VLANs and trunks, route between them, make the LAN resilient with STP, EtherChannel and HSRP, provide addresses with DHCPv4, SLAAC and DHCPv6, secure switch ports against layer 2 attacks, deploy a wireless LAN through a controller secured with WPA2, and compare it with WPA3, and configure and troubleshoot static routing. Every chapter ends with a lab in Cisco Packet Tracer, and the programme closes with a final project: a campus network built and presented by each participant. Enterprise Networking, Security & Automation (CCNA 3) follows.

At a glance

#	PHASE	FOCUS	HOURS
01	Switching & VLANs	Switch configuration, VLANs and trunks, inter-VLAN routing	9 h
02	Redundancy & Address Services	STP, EtherChannel, DHCPv4, SLAAC and DHCPv6, HSRP	9 h
03	LAN Security & Wireless	Layer 2 attack mitigation, WLAN with a controller, WPA2 and WPA3	6 h
04	Routing	Routing table, static and default routes, troubleshooting, skills integration	6 h
05	Final Project	A campus network, built and presented	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		42 h

Key facts

AUDIENCE	Participants who completed CCNA 1 or have equivalent networking experience
PREREQUISITES	Introduction to Networks (CCNA 1), or equivalent: IOS configuration, IPv4 and IPv6 addressing, subnetting. These topics are not taught again
DURATION	42 h: 36 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs on simulated Cisco devices
LAB ENVIRONMENT	Runs locally: one laptop per participant with Cisco Packet Tracer (free with each participant's Cisco Networking Academy account); physical Cisco switches, routers and a wireless controller for selected labs when the organiser provides a rack; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase and a final project presented to the group; certificate of completion for participants who complete both
OUTCOME	Participants can build a redundant, secured campus LAN with VLANs, inter-VLAN routing, wireless access and static routing, and troubleshoot it

Switching & VLANs

Participants configure switches for secure remote management, split the network into VLANs carried over 802.1Q trunks, then route between the VLANs with a router-on-a-stick and a layer 3 switch. Validated lab: three VLANs that reach each other through both inter-VLAN routing methods.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Switching Concepts & Configuration	Switch boot and SSH, frame forwarding, collision and broadcast domains	Lecture + Lab	3 h
1.2	VLANs & Trunking	VLANs, 802.1Q trunks, native VLAN, DTP	Lecture + Lab	3 h
1.3	Inter-VLAN Routing	Router-on-a-stick, layer 3 switch	Lecture + Lab	3 h

Redundancy & Address Services

Redundant links and gateways keep a LAN running when one part fails. Participants prevent loops with Rapid PVST+ and bundle links with EtherChannel, hand out IPv4 and IPv6 addresses with DHCPv4, SLAAC and DHCPv6, then protect the default gateway with HSRP. Validated lab: a LAN that keeps forwarding when a link or a gateway fails, with hosts addressed automatically in both IP versions.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	STP	Loop prevention, STP and Rapid PVST+, PortFast, BPDU guard	Lecture + Lab	3 h
2.2	EtherChannel	LACP and PAgP, configuration and verification	Lecture + Lab	1.5 h
2.3	DHCPv4	DHCPv4 server and relay	Lecture + Lab	1.5 h
2.4	SLAAC & DHCPv6	SLAAC, stateless and stateful DHCPv6	Lecture + Lab	1.5 h
2.5	First Hop Redundancy	FHRP concepts, HSRP	Lecture + Lab	1.5 h

PHASE 03

6 H

LAN Security & Wireless

Participants protect switch ports against common layer 2 attacks, then deploy a wireless LAN managed by a wireless LAN controller and secured with WPA2 and WPA3. Validated lab: a switch that blocks a rogue DHCP server with DHCP snooping and shuts down a port on a port-security violation, and a WLAN configured on the controller with WPA2-Personal and WPA2-Enterprise.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	LAN Security	Port security, DHCP snooping, dynamic ARP inspection, VLAN and STP attack mitigation	Lecture + Lab	3 h
3.2	Wireless LANs	802.11 concepts, WLAN components, WLC configuration, WPA2 and WPA3	Lecture + Lab	3 h

Routing

Participants read the routing table and configure IPv4 and IPv6 static, default and floating static routes, troubleshoot them methodically, then combine switching, routing and wireless in one integration lab. Validated lab: two sites joined by IPv4 and IPv6 static routes, with a floating static route that takes over when the primary link fails, inside the integrated switching and wireless topology.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Routing Concepts & Static Routing	Routing table, path selection, IPv4 and IPv6 static, default and floating static routes	Lecture + Lab	3 h
4.2	Troubleshooting Static Routes	Methodical troubleshooting of static and default routes	Lab	1.5 h
4.3	Skills Integration	Integrated switching, routing and wireless lab	Lab	1.5 h

PHASE 05

6 H

Final Project

Each participant builds a campus network in Packet Tracer from a written brief, reusing the phase labs, then documents it and defends the design in front of the group.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Brief & Build	Read the brief, then build VLANs and trunks, inter-VLAN routing, STP and EtherChannel, HSRP and static routing	Project	3 h
5.2	Services, Security & Documentation	DHCPv4 and DHCPv6, LAN security, the WLAN; topology diagram and addressing table	Project	1.5 h
5.3	Demo & Review	Failover demo (link and gateway), questions and feedback	Project	1.5 h

FINAL PROJECT

Campus Network

A campus network with VLANs and trunks, inter-VLAN routing, STP and EtherChannel, HSRP, DHCPv4 and DHCPv6, LAN security, a controller-based WLAN and static routing to the provider, which keeps working when a link or gateway fails.

STACK Cisco Packet Tracer, Cisco IOS switches and routers, wireless LAN controller, IPv4, IPv6

DELIVERABLE Packet Tracer file, topology diagram, addressing table, failover demo